BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The female reproductive system

The written notes for the female reproductive lecture: the ovary and its follicle stages, the uterine tubes, the uterus, the vagina and vulva, and the mammary glands. Module 5 competencies 146 to 150, with eight worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Before the competencies: the organs of the system

In your course notes: Module 5 packet, Chapter 3 introduction (page 21).

Dr. Rennie's video for this part: Female reproductive anatomy concept video, chapter Female reproductive system overview.

The female reproductive system makes oocytes, receives sperm, and supports a developing fetus. Six sets of organs do this work, and the five competencies in this chapter walk through them from the ovary outward.

Structure	What it is
Ovaries	The female gonads; they produce oocytes and secrete hormones
Uterine tubes	The fallopian tubes; they carry the oocyte toward the uterus
Uterus	The muscular organ where an embryo implants and a fetus develops
Vagina	The canal from the cervix to the outside; the birth canal
Vulva	The external genitalia
Mammary glands	The milk-producing glands of the breasts

- The order of the competencies follows the path an oocyte takes: ovary (146), uterine tube (147), uterus (148), vagina and vulva (149). The mammary glands (150) are last.
- A gonad is the organ that makes the gametes. In the female that is the ovary, and the gamete is the oocyte.

Competency 146. Ovary structure and follicle stages

In your course notes: Module 5 packet, Chapter 3, competency 146 (page 21).

Dr. Rennie's video for this part: Female reproductive anatomy concept video, chapter Ovary functions and hormones, and Ovary support ligaments.

You should be able to: Identify the ovary layers, ligaments, and follicle stages including the corpus luteum and corpus albicans.

The ovaries are the paired female gonads. Each is an almond-shaped organ against the lateral wall of the pelvic cavity, one on each side of the uterus, just below the open end of a uterine tube. The ovary produces oocytes and secretes hormones. To know it on a model or a slide you need three things: the ligaments that hold it, its layers from the outside in, and the follicle stages in its cortex.

The three ligaments of the ovary

The broad ligament is a large fold of peritoneum that drapes over the uterus, the uterine tubes, and the ovaries. A short fold of it, the mesovarium, attaches the ovary to the back of the broad ligament. The ovarian ligament is a cord of fibrous tissue that ties the ovary to the side of the uterus, just below where the uterine tube joins it. The suspensory ligament runs from the ovary up to the lateral pelvic wall, and it carries the ovarian artery and ovarian vein to and from the ovary. Like the testicular artery, the ovarian artery branches from the abdominal aorta.

Ligament	What it connects
Broad ligament (mesovarium)	Ovary to the back of the broad ligament, the fold of peritoneum over the uterus and tubes
Ovarian ligament	Ovary to the side of the uterus
Suspensory ligament	Ovary to the lateral pelvic wall; carries the ovarian artery and vein

- Memory tip: the suspensory ligament suspends the ovary from the pelvic wall and carries its vessels. The ovarian ligament ties the ovary to the uterus. These two get swapped a lot, so say the far end of each one out loud: suspensory to the wall, ovarian to the uterus.
- The mesovarium is not a separate ligament. It is the part of the broad ligament that holds the ovary.

The layers of the ovary, from the outside in

Germinal epithelium (ovarian surface epithelium)

A single layer of cuboidal cells covering the ovary. Despite its name, it does not make oocytes.

Tunica albuginea

A capsule of dense connective tissue under the epithelium. It has the same name as the capsule of the testis.

Ovarian cortex

The outer region of the ovary proper. It holds the follicles, so every follicle stage you name on a slide is in the cortex.

Ovarian medulla

The center: loose connective tissue with the blood vessels, lymphatic vessels, and nerves.

The follicle stages

An ovarian follicle is an oocyte plus the cells around it. The cells next to the oocyte are follicle cells, called granulosa cells once they become cuboidal. In later stages, connective tissue cells outside them form a layer called the theca. Follicles grow through a set series of stages, so on a slide you name each one by its size, its layers of cells, and whether it has a fluid space.

A primordial follicle is a primary oocyte with one layer of flat follicle cells. These are the smallest and most numerous follicles, in the outer cortex. In a primary follicle, the follicle cells become cuboidal granulosa cells and a clear layer, the zona pellucida, forms around the oocyte. A secondary follicle has several layers of granulosa cells, a theca around the outside, and small pockets of fluid between the granulosa cells. A mature follicle (vesicular or Graafian follicle) has one large fluid-filled cavity, the antrum. Its oocyte sits on a mound of granulosa cells to one side, ringed by the corona radiata. It is big enough to bulge from the surface of the ovary. Just before ovulation, its oocyte becomes a secondary oocyte.

At ovulation, the mature follicle ruptures at the ovary surface and releases the secondary oocyte with its corona radiata into the pelvic cavity. The granulosa and theca cells left behind become the corpus luteum (yellow body), a large glandular body that secretes hormones. If no pregnancy occurs, the corpus luteum breaks down into the corpus albicans (white body), a pale scar of fibrous tissue. Fluid-filled ovarian cysts often form from a follicle or a corpus luteum that does not go away.

Stage	Oocyte	Cells around it	How to spot it
Primordial follicle	Primary oocyte	One layer of flat follicle cells	Smallest; many, in the outer cortex
Primary follicle	Primary oocyte	Cuboidal granulosa cells; zona pellucida forms	Oocyte ringed by cuboidal cells
Secondary follicle	Primary oocyte	Several layers of granulosa cells and a theca	Small fluid pockets between the granulosa cells
Mature follicle	Primary oocyte; it becomes a secondary oocyte just before ovulation	Granulosa cells, corona radiata, theca	One large antrum; bulges from the ovary
Corpus luteum	None (released at ovulation)	Granulosa and theca cells left behind	Large, pale or yellow glandular body
Corpus albicans	None	Fibrous scar tissue	Pale, white scar in the cortex

- The order is fixed: primordial, primary, secondary, mature follicle, then ovulation, then corpus luteum, then corpus albicans. The corpus luteum forms after ovulation, from what the follicle left behind, so it never comes before the oocyte is released.
- Primordial and primary follicles both hold a primary oocyte. What changes is the follicle cells: flat in the primordial follicle, cuboidal in the primary follicle, with the zona pellucida appearing at the primary stage.
- The mature follicle is the only stage with one big antrum. A secondary follicle has fluid, but only in small pockets.
- Luteum means yellow and albicans means white. The yellow body is the working gland; the white body is the scar it leaves.

Competency 147. Uterine tubes and ovum pathway

In your course notes: Module 5 packet, Chapter 3, competency 147 (page 23).

Dr. Rennie's video for this part: Female reproductive anatomy concept video, chapter Uterine tubes: role and layers.

You should be able to: Trace the oocyte from the ovary to the uterus, naming the fimbriae, infundibulum, ampulla, and isthmus.

The uterine tubes (fallopian tubes) are a pair of tubes about 10 cm long. Each runs from an upper corner of the uterus out toward an ovary, inside the upper edge of the broad ligament. This part of the broad ligament is the mesosalpinx. The tube carries the oocyte toward the uterus, and it is where fertilization usually happens.

The outer end of the tube is not attached to the ovary. At ovulation the oocyte is released into the pelvic cavity, and the tube has to catch it. The tube has four named regions, from the ovary toward the uterus.

The regions of the tube, from the ovary to the uterus

The infundibulum is the funnel-shaped outer end, open to the pelvic cavity near the ovary. Its edge is fringed with fimbriae, finger-like projections that drape over the ovary. Around ovulation the fimbriae sweep across the surface of the ovary and draw the oocyte into the funnel. The ampulla is the widest and longest part of the tube. Fertilization usually happens here. The isthmus is the narrow, thick-walled part that joins the uterus. A short intramural part then passes through the uterine wall and opens into the uterine cavity.

Region	What it is
Fimbriae	Finger-like projections on the edge of the infundibulum that sweep the oocyte into the tube
Infundibulum	The funnel-shaped outer end, open to the pelvic cavity near the ovary
Ampulla	The widest and longest part; the usual site of fertilization
Isthmus	The narrow, thick-walled part next to the uterus
Intramural part	The short part that passes through the uterine wall into the uterine cavity

- Memory tip: the tube narrows as it nears the uterus: fimbriae and infundibulum (the open funnel), then the wide ampulla, then the narrow isthmus. An isthmus of land is a narrow neck, and so is this one.
- The ampulla is the wide part and the isthmus is the narrow part next to the uterus. Students flip these two more than anything else in this competency.

What moves the oocyte

The lining of the tube has ciliated cells and secretory cells. The cilia beat toward the uterus, and a layer of smooth muscle in the wall squeezes in waves. Together they move the oocyte, or the early embryo, toward the uterus over a few days. On a drawing, the cilia arrow always points toward the uterus, never toward the ovary.

The route from the ovary to the uterus

1. Mature follicle in the ovarian cortex: the follicle ruptures at ovulation and releases the secondary oocyte into the pelvic cavity.
2. Fimbriae: the fimbriae sweep over the ovary and draw the oocyte in.
3. Infundibulum: the funnel carries the oocyte into the tube.
4. Ampulla: the oocyte moves through the widest part of the tube, where sperm usually meet and fertilize it.
5. Isthmus: the oocyte or early embryo moves through the narrow part of the tube.
6. Intramural part: it passes through the uterine wall.
7. Uterine cavity: it arrives in the cavity of the uterus. An embryo implants in the endometrium, usually in the upper part of the body of the uterus.

An open pathway to the pelvic cavity

The female tract is open to the pelvic cavity. The infundibulum opens near the ovary and does not seal onto it. That opening is how the fimbriae catch an oocyte. It is also how an infection can climb from the vagina through the cervix, uterus, and uterine tubes into the pelvic cavity. This spreading infection is pelvic inflammatory disease (PID).

In an ectopic pregnancy, an embryo implants outside the uterine cavity. Most implant in the uterine tube, usually in the ampulla. This happens when the embryo is held up inside the tube, for example where scarring from earlier PID narrows it. The thin wall of the tube cannot stretch to hold a growing embryo. In rare cases an embryo implants in the pelvic or abdominal cavity, which is possible because the infundibulum is open.

What this means for patients

- Fertilization usually happens in the ampulla, the wide part of the tube, not in the uterus. The embryo then travels the rest of the tube and implants in the endometrium.
- Because the infundibulum is open, an infection that starts in the vagina has a continuous route up through the cervix, uterus, and tubes into the pelvic cavity. That is pelvic inflammatory disease.
- Scarring from PID can narrow the tube and hold up an embryo, which is one reason ectopic pregnancies most often happen in the ampulla.

Competency 148. Uterus regions, ligaments, and wall

In your course notes: Module 5 packet, Chapter 3, competency 148 (page 24).

Dr. Rennie's video for this part: Female reproductive anatomy concept video, chapter Uterus regions and support, and Cervix mucus and purpose.

You should be able to: Name the regions and supporting ligaments of the uterus and identify the perimetrium, myometrium, and endometrium.

The uterus is a hollow, thick-walled, pear-shaped organ in the pelvic cavity. It sits behind the urinary bladder and in front of the rectum. In most women it tips forward over the bladder. It receives the embryo, holds the developing fetus, and contracts during childbirth.

Regions

The fundus is the rounded top, above the points where the uterine tubes enter. The body is the main central part. The space inside it is the uterine cavity. Below the body, a short narrowed region called the isthmus of the uterus leads to the cervix, the narrow lower part that projects into the vagina. The inside of the cervix is the cervical canal. It opens into the uterine cavity at the internal os and into the vagina at the external os.

Region	What it is
Fundus	The rounded top, above the entry of the uterine tubes
Body	The main central part; holds the uterine cavity
Cervix	The narrow lower part that projects into the vagina
Uterine cavity	The space inside the body of the uterus
Cervical canal	The channel through the cervix, from the internal os to the external os

- Os means mouth. The internal os is the inner mouth of the cervical canal, opening into the uterine cavity. The external os is the outer mouth, opening into the vagina. Do not flip them.
- The uterus has an isthmus and so does the uterine tube. Both are narrow necks: the isthmus of the uterus sits between the body and the cervix, and the isthmus of the tube sits next to the uterus.

Supporting ligaments

Four paired ligaments hold the uterus, with help from the muscles of the pelvic floor. The broad ligaments are a double fold of peritoneum from the sides of the uterus to the lateral pelvic walls. The part of it beside the uterus is the mesometrium; the same sheet also holds the uterine tubes (mesosalpinx) and the ovaries (mesovarium). The round ligaments are fibrous cords that leave the uterus just below each uterine tube, run forward through the inguinal canal, and end in the labia majora. They help keep the uterus tipped forward. The cardinal ligaments (transverse cervical ligaments) run from the cervix and upper vagina to the lateral pelvic walls. They are the main support that keeps the uterus from sliding down. The uterosacral ligaments run from the cervix back to the sacrum.

Ligament	Runs from	Runs to
Broad ligaments	Sides of the uterus	Lateral pelvic walls and floor
Round ligaments	Uterus, just below each uterine tube	Through the inguinal canal to the labia majora
Cardinal ligaments	Cervix and upper vagina	Lateral pelvic walls
Uterosacral ligaments	Cervix	Sacrum

- Memory tip: the round ligament takes the same route through the inguinal canal that the spermatic cord takes in a male.
- One broad ligament, three named parts: mesometrium beside the uterus, mesosalpinx holding the tube, mesovarium holding the ovary.
- Two ligaments leave from the cervix. The uterosacral goes back to the sacrum. The cardinal goes sideways to the pelvic wall and is the main thing keeping the uterus from dropping.

The three layers of the wall, from the outside in

The perimetrium is the outer serous layer. It is part of the visceral peritoneum. The myometrium is the thick middle layer of smooth muscle, thickest in the fundus. It contracts during childbirth. The endometrium is the inner mucous lining, where an embryo implants. It has two layers. The stratum functionalis (functional layer), next to the cavity, is shed during menstruation. The stratum basalis (basal layer), next to the myometrium, stays in place and rebuilds the functional layer each cycle.

Layer	What it is
Perimetrium	The outer serous layer; part of the visceral peritoneum
Myometrium	The thick middle layer of smooth muscle that contracts during childbirth
Endometrium	The inner lining; its stratum functionalis sheds in menstruation and its stratum basalis rebuilds it

- Peri means around, myo means muscle, endo means inside. Perimetrium outside, myometrium in the middle, endometrium lining the cavity.
- The layer that contracts during childbirth is the myometrium, the muscle layer, not the endometrium.
- Inside the endometrium, the functional layer is the one next to the cavity and the one that leaves each month. The basal layer stays and regrows it.

What this means for patients

Three common disorders each belong to one part of the uterus.

Disorder	Part of the uterus	What it is
Uterine fibroids	Myometrium	Non-cancerous tumors of the smooth muscle of the myometrium
Endometriosis	Endometrium	Endometrial tissue grows outside the uterus, for example on the ovaries or the pelvic peritoneum
Cervical cancer	Cervix	A malignant tumor of the cervix

- Fibroids are muscle, so they live in the myometrium. Endometriosis is lining tissue in the wrong place, so it is the endometrium gone outside the uterus.

Competency 149. Vagina and external genitalia

In your course notes: Module 5 packet, Chapter 3, competency 149 (page 26).

Dr. Rennie's video for this part: Female reproductive anatomy concept video, chapter Vagina structure and protection, and External genitalia and breasts.

You should be able to: Identify the vagina, fornix, hymen, and the vulvar structures including the clitoris, vestibule, labia, and greater vestibular glands.

The vagina

The vagina is a thin-walled fibromuscular tube about 10 cm long. It runs from the cervix down and forward to the outside. It lies behind the bladder and urethra and in front of the rectum. It receives the penis, carries menstrual flow out, and serves as the birth canal. Its lining is stratified squamous epithelium in folds called rugae, with a layer of smooth muscle around it. It opens to the outside at the vaginal orifice.

The cervix pokes into the top of the vagina. The recess that rings the cervix is the fornix. Its posterior part, the posterior fornix, is the deepest. The hymen is a thin fold of mucous membrane at the vaginal orifice that partly covers it. Its shape and size vary a great deal from person to person.

- Front to back in the pelvis: bladder and urethra, then the vagina, then the rectum. The vagina is the middle of the three.
- The fornix is at the top of the vagina, around the cervix. The hymen is at the bottom, at the opening. They are the two ends of the same tube.

The vulva

The external genitalia together are the vulva. The mons pubis is a rounded pad of fat over the pubic symphysis. The labia majora are two folds of skin and fat that run back from the mons pubis. They are covered with hair after puberty. The labia minora are two thin, hairless folds inside the labia majora. At the front, they meet over the clitoris and form its prepuce.

The clitoris is a small erectile organ at the front of the vulva, where the labia minora meet. Like the penis, it is

built of erectile tissue, and only its tip, the glans of the clitoris, shows on the surface. The vestibule is the space between the labia minora. It holds the two main openings: the external urethral orifice toward the front and the vaginal orifice behind it.

The greater vestibular glands (Bartholin glands) are a pair of pea-sized glands, one on each side of the vaginal orifice, slightly toward the back. Their ducts open into the vestibule, and their mucus lubricates the vestibule. The perineum is the diamond-shaped region between the pubic symphysis and the coccyx. Its front half, the urogenital triangle, holds the external genitalia. Its back half, the anal triangle, holds the anus.

Structure	Where it is	What it is
Vagina	From the cervix to the vestibule	Fibromuscular canal; the birth canal
Fornix	Top of the vagina, around the cervix	Recess that rings the cervix
Hymen	At the vaginal orifice	Thin fold of mucous membrane that partly covers the opening
Mons pubis	Over the pubic symphysis	Rounded pad of fat
Labia majora	Outer folds, back from the mons pubis	Two folds of skin and fat, hair covered after puberty
Labia minora	Inside the labia majora	Two thin, hairless folds that enclose the vestibule
Clitoris	Front of the vulva, where the labia minora meet	Small erectile organ
Vestibule	Between the labia minora	Space holding the urethral and vaginal openings
Greater vestibular glands	On each side of the vaginal orifice	Glands whose mucus lubricates the vestibule
Perineum	Between the pubic symphysis and the coccyx	Diamond-shaped region with the urogenital and anal triangles

- Memory tip: front to back along the midline: mons pubis, clitoris, external urethral orifice, vaginal orifice, then the anus behind the vestibule. A urethral opening drawn behind the vaginal opening is the usual slip.
- The vestibule is the area between the labia minora, not between the labia majora. Majora means larger; minora means smaller and sits inside.

Female structures and their male matches

Several female structures develop from the same embryonic tissue as a male structure, so they are homologous: the same starting material built into a different shape.

Female	Male
Ovaries	Testes
Labia majora	Scrotum
Clitoris	Penis
Greater vestibular glands	Bulbourethral glands

- The three vulvar homologs to know: labia majora match the scrotum, the clitoris matches the penis, and the greater vestibular glands match the bulbourethral glands. Both gland pairs are pea-sized and both secrete lubricating mucus.

Competency 150. Mammary gland structure

In your course notes: Module 5 packet, Chapter 3, competency 150 (page 27).

Dr. Rennie's video for this part: Female reproductive anatomy concept video, chapter External genitalia and breasts.

You should be able to: Identify the lobes, lobules, alveoli, lactiferous ducts, nipple, areola, and suspensory ligaments of the mammary gland.

The mammary glands are modified sweat glands in the breasts. Each breast lies in the subcutaneous tissue in front of the pectoralis major muscle. Much of the breast is adipose tissue, and the amount of fat mostly sets the size of the breast.

Nested units: lobes, lobules, alveoli

The gland is built in nested units. It has 15 to 20 lobes arranged in a ring around the nipple, separated by fat. Each lobe divides into smaller lobules. Each lobule holds clusters of alveoli, small sacs lined with cells that secrete milk. Myoepithelial cells around the alveoli contract to push the milk into the ducts.

Ducts, nipple, and areola

Small ducts from the alveoli join into one lactiferous duct for each lobe. Just before the nipple, each lactiferous duct widens into a lactiferous sinus, where milk collects. The ducts open on the nipple, the raised projection near the center of the breast. The areola is the ring of pigmented skin around the nipple. Its small bumps are areolar glands, a type of sebaceous gland.

Suspensory ligaments

The suspensory ligaments (Cooper ligaments) are bands of connective tissue that run from the skin through the breast, between the lobes, to the fascia over the pectoralis major. They support the breast.

Structure	What it is
Lobes	The 15 to 20 compartments of the gland, each divided into lobules
Lobules	The subdivisions of a lobe, each holding clusters of alveoli
Alveoli	The small sacs of gland cells that secrete milk
Lactiferous ducts	One duct per lobe; carries milk toward the nipple
Lactiferous sinus	The widened end of a lactiferous duct where milk collects
Nipple	The projection where the lactiferous ducts open
Areola	The ring of pigmented skin around the nipple
Suspensory ligaments	Bands of connective tissue from the skin to the fascia over the pectoralis major that support the breast

The route of milk, from where it is made to the outside

1. Alveoli: gland cells inside each lobule secrete milk.
2. Lobule ducts: small ducts carry milk out of each lobule.
3. Lactiferous duct: one duct drains each lobe toward the nipple.
4. Lactiferous sinus: the duct widens and milk collects.
5. Nipple: milk leaves through openings on the nipple.

- Memory tip: the gland nests one way: alveoli inside lobules, lobules inside lobes. Each lobe drains by its own lactiferous duct to the nipple.
- The alveoli are the smallest sacs and the lobe is the biggest unit. Small sits inside big.
- The lactiferous ducts open at the nipple, not on the areola. The areola has its own small sebaceous glands, the areolar glands, which are not milk ducts.
- Breast alveoli and lung alveoli share a name because both are small sacs. The ones here make milk.

What this means for patients

- Most breast cancer starts in the ducts.
- A tumor that pulls on a suspensory ligament can dimple the skin over it, because the ligament runs all the way from the skin down to the fascia over the pectoralis major.

Worked questions

Eight questions in the style of the lecture exam, covering every competency. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 146

Which ligament ties the ovary to the lateral pelvic wall and carries the ovarian artery and vein?

A. Ovarian ligament

B. Mesovarium

C. Suspensory ligament

D. Round ligament

Answer: C. The suspensory ligament runs from the ovary up to the lateral pelvic wall, and the ovarian artery and vein travel inside it to reach the ovary. Suspensory suspends the ovary from the wall.

Why the others are wrong

- **A** The ovarian ligament is a fibrous cord that ties the ovary to the side of the uterus, not to the pelvic wall, and it carries no vessels.
- **B** The mesovarium is the short fold of the broad ligament that attaches the ovary to the back of the broad ligament. It is a peritoneal fold, not the route of the ovarian vessels.
- **D** The round ligament belongs to the uterus. It leaves the uterus below the tube, runs through the inguinal canal, and ends in the labia majora.

QUESTION 2 · COMPETENCY 146

On an ovary slide you see a large, pale yellow glandular body in the cortex with no oocyte inside. What is it?

A. Corpus luteum

B. Mature follicle

C. Corpus albicans

D. Secondary follicle

Answer: A. The corpus luteum (yellow body) is what the granulosa and theca cells become after the mature follicle ruptures at ovulation. It is a large glandular body that secretes hormones, and it has no oocyte because the oocyte was released.

Why the others are wrong

- **B** A mature follicle still holds its oocyte, sitting on a mound of granulosa cells beside one large antrum. It is a follicle with fluid, not a solid gland.
- **C** The corpus albicans (white body) is the pale, fibrous scar left when the corpus luteum breaks down. It is scar tissue, not a glandular body.
- **D** A secondary follicle holds a primary oocyte inside several layers of granulosa cells with small fluid pockets. It has an oocyte and is much smaller.

QUESTION 3 · COMPETENCY 147

Where in the uterine tube does fertilization usually happen?

A. Infundibulum

B. Isthmus

C. Intramural part

D. Ampulla

Answer: D. The ampulla is the widest and longest part of the tube, and it is the usual site of fertilization. The oocyte spends enough time there for sperm to reach it before it moves on toward the uterus.

Why the others are wrong

- **A** The infundibulum is the funnel-shaped open end that catches the oocyte with its fimbriae. The oocyte enters the tube here but is not usually fertilized here.
- **B** The isthmus is the narrow, thick-walled part next to the uterus. By the time the oocyte or embryo reaches it, fertilization has usually already happened.
- **C** The intramural part is the short segment passing through the uterine wall into the uterine cavity, the last stretch of the tube.

QUESTION 4 · COMPETENCY 147

Which structure explains how an infection can climb from the vagina all the way into the pelvic cavity?

A. The closed end of the uterine tube

B. The open infundibulum of the uterine tube

C. The internal os of the cervix

D. The posterior fornix of the vagina

Answer: B. The infundibulum opens into the pelvic cavity near the ovary and does not seal onto it. This is how the fimbriae catch an oocyte, and it is also the route an infection takes from the vagina through the cervix, uterus, and tubes into the pelvic cavity. That spreading infection is pelvic inflammatory disease.

Why the others are wrong

- **A** The uterine tube has no closed end. It opens into the uterus at one end and into the pelvic cavity at the other, which is exactly the problem.
- **C** The internal os is the opening of the cervical canal into the uterine cavity. An infection passes through it on the way up, but it is the open tube, not the os, that lets infection leave the tract and reach the pelvic cavity.
- **D** The posterior fornix is the deepest part of the recess around the cervix at the top of the vagina. It is a dead-end pocket, not an opening into the pelvic cavity.

QUESTION 5 · COMPETENCY 148

Which layer of the uterine wall contracts during childbirth?

A. Myometrium

B. Endometrium

C. Perimetrium

D. Stratum basalis

Answer: A. The myometrium is the thick middle layer of smooth muscle, thickest in the fundus. Myo means muscle, and the muscle layer is the one that contracts.

Why the others are wrong

- **B** The endometrium is the inner mucous lining where an embryo implants. Its functional layer sheds in menstruation. It is lining, not muscle.
- **C** The perimetrium is the thin outer serous layer, part of the visceral peritoneum. It has no muscle to contract.
- **D** The stratum basalis is the deeper layer of the endometrium that rebuilds the functional layer each cycle. It is part of the lining, not of the muscle.

QUESTION 6 · COMPETENCY 148

Which ligament leaves the uterus just below each uterine tube, passes through the inguinal canal, and ends in the labia majora?

A. Cardinal ligament

B. Uterosacral ligament

C. Round ligament

D. Broad ligament

Answer: C. The round ligaments are fibrous cords that take the same route through the inguinal canal that the spermatic cord takes in a male, and they end in the labia majora. They help keep the uterus tipped forward over the bladder.

Why the others are wrong

- **A** The cardinal (transverse cervical) ligaments run from the cervix and upper vagina sideways to the lateral pelvic walls. They are the main support that keeps the uterus from sliding down.
- **B** The uterosacral ligaments run from the cervix back to the sacrum.
- **D** The broad ligaments are the double fold of peritoneum from the sides of the uterus to the lateral pelvic walls. They are a sheet, not a cord, and they never enter the inguinal canal.

QUESTION 7 · COMPETENCY 149

The vestibule of the vulva is the space between the

A. labia majora

B. labia minora

C. clitoris and the mons pubis

D. vaginal orifice and the anus

Answer: B. The vestibule is the space enclosed by the two labia minora. It holds the external urethral orifice toward the front and the vaginal orifice behind it, and the ducts of the greater vestibular glands open into it.

Why the others are wrong

- **A** The labia majora are the larger, hair-covered outer folds. The labia minora sit inside them, and the vestibule is inside the labia minora, not between the majora.
- **C** The mons pubis is the fat pad over the pubic symphysis, and the clitoris sits just behind it where the labia minora meet. There is no named space between them; the vestibule is behind the clitoris.
- **D** The region behind the vaginal orifice toward the anus is part of the perineum, not the vestibule.

QUESTION 8 · COMPETENCY 150

Milk leaving an alveolus follows which route to the outside?

A. Lactiferous sinus, lactiferous duct, lobule ducts, nipple

B. Lobule ducts, lactiferous sinus, lactiferous duct, areola

C. Lactiferous duct, lobule ducts, lactiferous sinus, nipple

D. Lobule ducts, lactiferous duct, lactiferous sinus, nipple

Answer: D. Milk made in the alveoli drains through the small ducts of the lobule into the one lactiferous duct for that lobe. Just before the nipple the duct widens into the lactiferous sinus, where milk collects, and then it opens on the nipple.

Why the others are wrong

- **A** This runs the path backward. The sinus is the widened end of the lactiferous duct just before the nipple, so it comes after the duct, not before the lobule ducts.
- **B** The sinus comes after the lactiferous duct, not before it, and the ducts open on the nipple, not on the areola. The areola is the pigmented ring around the nipple with its own small sebaceous glands.
- **C** The lobule ducts are the small ducts that feed into the lactiferous duct, so they must come first. The lactiferous duct collects from the whole lobe.